

Bluetooth based DC/BLDC motor speed control (clockwise /anticlockwise) by using STM32-NUCLEO-G431RB

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Abstract—In the modern world, there are many different uses for DC/BLDC motor speed control. This study proposes a method for Bluetooth-based wireless speed as well as direction control of DC/BLDC motors. The Bluetooth module, that is attached to the STM32 G431 board, receives the signal provided by the Android mobile device. The DC Motor's speed is managed using the Pulse Width Modulation (PWM) technology. Different signals are created to control MCU to transmit PWM signals and direction control signals to motor driver, which may perform operations like speed altering and direction alternating, in accordance with various host mobile APP commands (clockwise and anticlockwise and Speed control). Index Terms: Bluetooth Module, Motor Driver (L298N 2A), STM32-NUCLEO-G431RB

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I. INTRODUCTION

A direct current (DC) motor is a permanent magnet synchronous electric motor. Unlike standard DC motors, which use brushes to physically make contact with the commutator on the rotor and complete the electrical route from the power source to the armature windings. There are no brushes in BLDC motors; instead, a permanent magnet rotor and a stator with a series of coils are used to commutate the electrical current. D.C. motor speed control is crucial for industrial automation in the production process. The actuators for industrial robots, drones, fans, and CD/DVD drives are just a few of the many uses for DC motors. Recent technological advancements have made wireless or remote control of motors popular because of how simple it is to use and operate from a distance. The suggested work's goal is to create a prototype for a potential application. Here, a Bluetooth module and a DC motor, two separate technologies, are combined. We can make the control cordless thanks to Bluetooth technology.

II. DESIGN AND IMPLEMENTATION

A. Hardware Design

Figure 1 shows the block diagram for a DC motor's speed and direction control. The Android application transmits this signal, which is then sent to the Bluetooth module and received by the STM32 G431 Microcontroller. The motor may be driven either clockwise or anticlockwise using this

system. Reversing the current flow direction in the Brushless DC motor's Armature or Field winding is how the motor's direction is controlled.

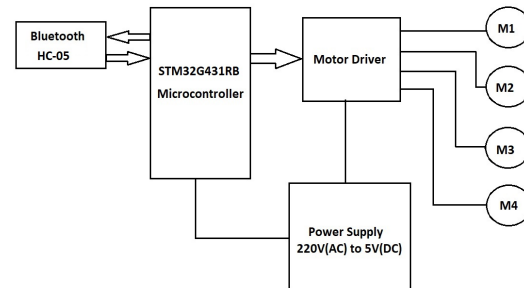


Fig. 1. Block Diagram

System Schematic for Proposed Speed Control as seen in Fig. 2. The user uses an Android smartphone that is wirelessly linked with a Bluetooth module to give commands to the STM32 G431 microcontroller. The STM32 G431 microcontroller creates the direction and speed control signal and transmits it to the motor driver once the Bluetooth module receives the data given by the user.

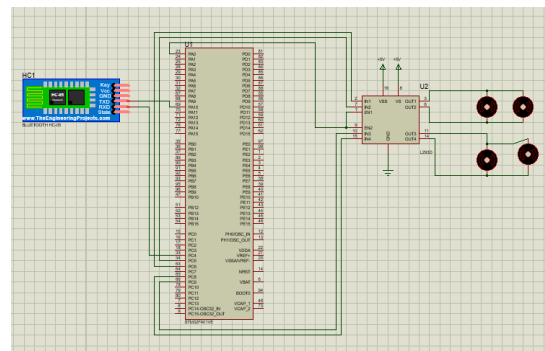


Fig. 2. Schematic Diagram

III. PWM TECHNIQUE

The most used technique for managing speed is PWM. It is more effective and simple to put into practise. Over 1KHz is the frequency at which it produces a square wave. By altering the duty cycle, the DC motor's speed may be managed..

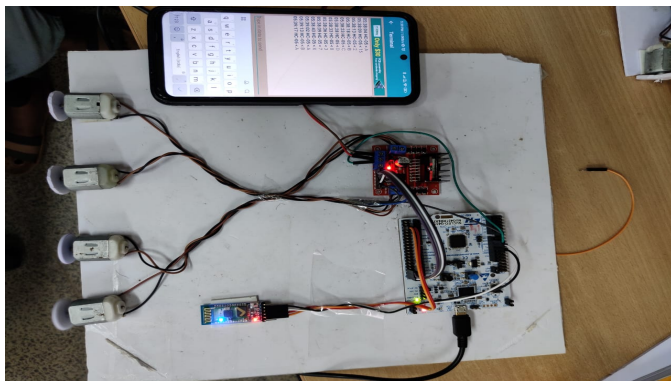


Fig. 3. Speed and DirectionControl

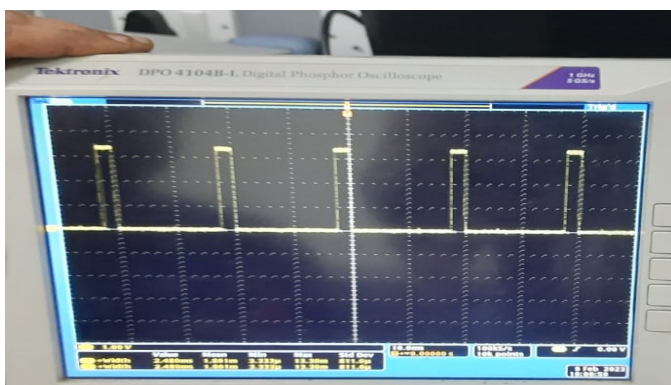


Fig. 4. PWM Waveform

Duty Cycle = $(\text{Ton}/(\text{Ton}+\text{Toff})) \times 100$, where Ton signifies the period of time during which the square wave is at its highest. Toff stands for the duration of the square wave's low period. Total time is the interval (Ton+Toff). Typically, the duty cycle is stated as a percentage. When the duty cycle is increased, the speed also rises, and vice versa. PWM waveform is shown in Fig 4 and Fig 7.

IV. HARDWARE COMPONENTS

The Nucleo-G431RB microcontroller, Motor Driver, Bluetooth Module, and DC motors are just a few of the parts that make up the system.

A. Microcontroller

STM32 G4 family of microcontrollers features a wide range of parts with specific peripherals appropriate for various applications and is developed for ultra-low power applications. These controllers' design is especially well-suited for mobile

measuring equipment, with five low-power modes being included to extend battery life.

The microcontroller has an effective 16-bit register system, a potent 32-bit RISC CPU, and built-in constant generators for the best possible code execution. Microcontroller is shown in Fig 5.

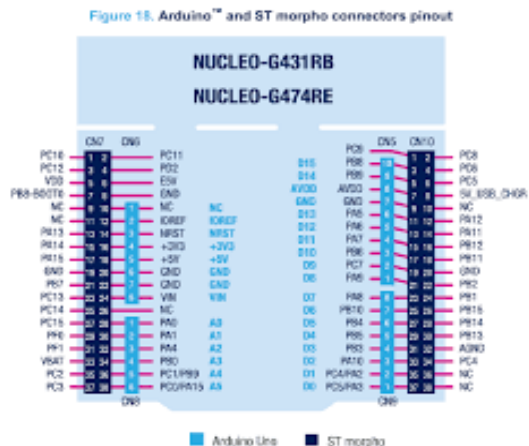


Fig. 5. Microcontroller

B. Bluetooth Module HC05

In this work, the HC05 module is used for Bluetooth connectivity. For a wireless, transparent communication configuration, it is necessary. A preferable option for wireless communication is to use the module's Master/Slave setup. It is a Bluetooth V2.0 module with EDR. Additionally, it contains a baseband and 2.4 GHz transceiver. Fig 6 shows the bluetooth Module.

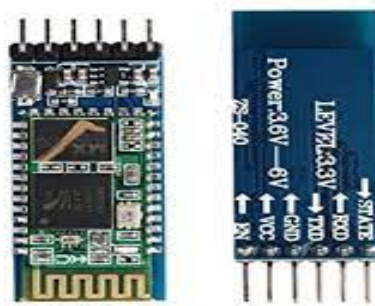


Fig. 6. Bluetooth Module

C. DC Motor

Fig. 7 shows the DC Motor. It consumes 250mA and work at 6volts

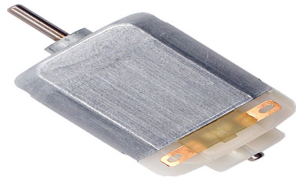


Fig. 7. Motor

D. Motor Driver

Motor driver is a module which is used to control the speed and direction of DC motors and Brushless DC motors. In this module one Voltage regulator is there which converts the input voltage into 5V. In this module 2 H-Bridge is there in L298 IC. So this motor driver has 2 channel and we can control 2 DC motors. IN1, IN2 pins are used to control the direction of one channel (Motor 1) and ENA pin is used to give PWM signal to control the speed. Similarly IN3 and IN4 pins are used to control the direction for channel 2 (Motor 2) and ENB is used to give PWM signal to control the speed. This motor Driver can deliver maximum 25 Watts Power. Fig 9 shows the internal circuit of Motor Driver. Fig 4 and fig 8 shows the PWM signal which we are applying at ENA and ENB of motor driver. By changing the duty cycle we can control the speed of Motors.

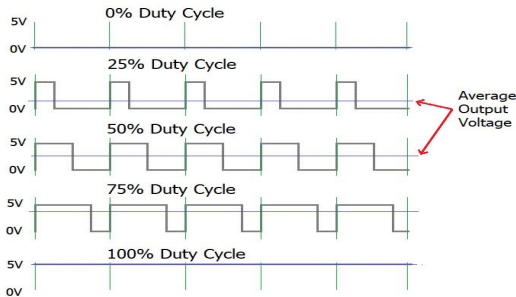


Fig. 8. PWM Waveform

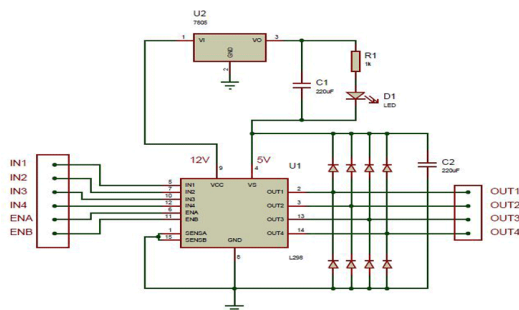


Fig. 9. Motor Driver

V. FLOW CHART

Fig 10 shows the Flow chart of the project.

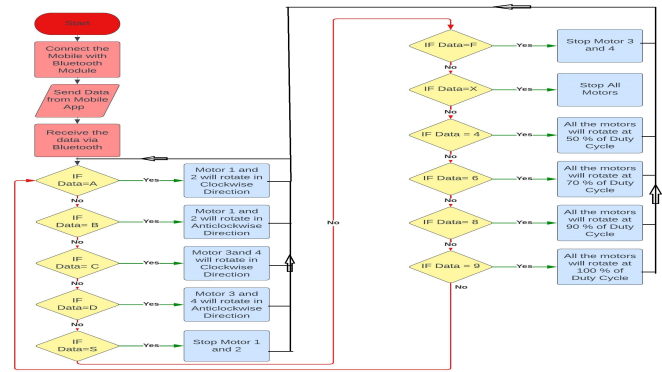


Fig. 10. Flow Chart

VI. RESULT

With this suggested setup, the various speeds with forward and reverse rotational directions are attained. The Speed as well as Direction control of Motors using Bluetooth by utilising STM32-NUCLEO-G431RB project was created to control the speed and direction of a motor using Bluetooth. As we can regulate the speed of the motor through utilising the PWM signal.

VII. CONCLUSION

In this work, Bluetooth and STM32-NUCLEO-G431RB has been used to control the speed as well as direction of a motor. As the duty cycle varied, terminal voltage did as well. The circuit utilised in the suggested scheme is simple to use and might be beneficial for a number of remote motor speed control applications.

VIII. REFERENCES

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